

PROGRAM

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, head, max_disk, total_seek = 0;

    printf("Enter max disk size: ");
    scanf("%d", &max_disk);
    if (max_disk < 0) { printf("Invalid Input\n"); exit(0); }

    printf("Enter initial head position: ");
    scanf("%d", &head);
    if (head < 0) { printf("Invalid Input\n"); exit(0); }

    printf("Enter number of requests: ");
    scanf("%d", &n);

    int req[n + 2];
    printf("Enter the requests: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &req[i]);
        if (req[i] < 0 || req[i] > max_disk) { printf("Invalid Input\n"); exit(0); }
    }

    // Add head and max_disk end for SCAN (Assuming initial direction is Right)
    req[n] = head;
    req[n+1] = max_disk;
    n = n + 2;

    // Sort the array
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (req[j] > req[j + 1]) {
                int temp = req[j]; req[j] = req[j + 1]; req[j + 1] = temp;
            }
        }
    }

    int pos;
    for (int i = 0; i < n; i++) {
        if (req[i] == head) { pos = i; break; }
    }

    // Move Right
    for (int i = pos; i < n; i++) {
        total_seek += abs(head - req[i]);
        head = req[i];
    }

    // Reverse Direction (Move Left)
    for (int i = pos - 1; i >= 0; i--) {
        total_seek += abs(head - req[i]);
        head = req[i];
    }

    printf("Total Seek Operations (SCAN) = %d\n", total_seek);
}
```

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    return 0;  
}
```

OUTPUT

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_10$ gcc scan.c  
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_10$ ./a.out  
Enter max disk size: 199  
Enter initial head position: 53  
Enter number of requests: 8  
Enter the requests: 98  
183  
41  
122  
14  
124  
65  
67  
Total Seek Operations (SCAN) = 331  
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_10$
```